

Automatic Parking Barrier System

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1 Project Description

The goal of this project was to create an automatic parking barrier system in Tinkercad. The system detects a vehicle approaching the entrance, opens the barrier automatically, detects when the vehicle has passed through, and then closes the barrier again.

The system uses two HC-SR04 ultrasonic sensors and one servo motor. The first ultrasonic sensor detects a vehicle approaching the barrier. If the vehicle is within 100 cm, the servo motor moves the barrier from 90 degrees to 0 degrees and opens it. The second ultrasonic sensor is used after the vehicle has entered. When the vehicle is detected by the second sensor and later moves more than 100 cm away from it, the barrier closes again.

Three LEDs are used to show the current state of the system. The red LED is on when the barrier is closed, the yellow LED is on while the barrier is moving, and the green LED is on when the barrier is fully open.

Parts used:

- Arduino Uno
- 2 HC-SR04 ultrasonic sensors
- Servo motor
- Red, yellow and green LEDs
- 3 220 Ω resistors
- Breadboard
- Jumper wires

Wiring: The first ultrasonic sensor uses pins 7 and 6 for TRIG and ECHO. The second ultrasonic sensor uses pins 3 and 2. The servo motor is connected to pin 4. The green LED is connected to pin 8, the yellow LED to pin 12, and the red LED to pin 13. Each LED is connected through a 220 Ω resistor. The breadboard is used to distribute 5V and GND to the components.

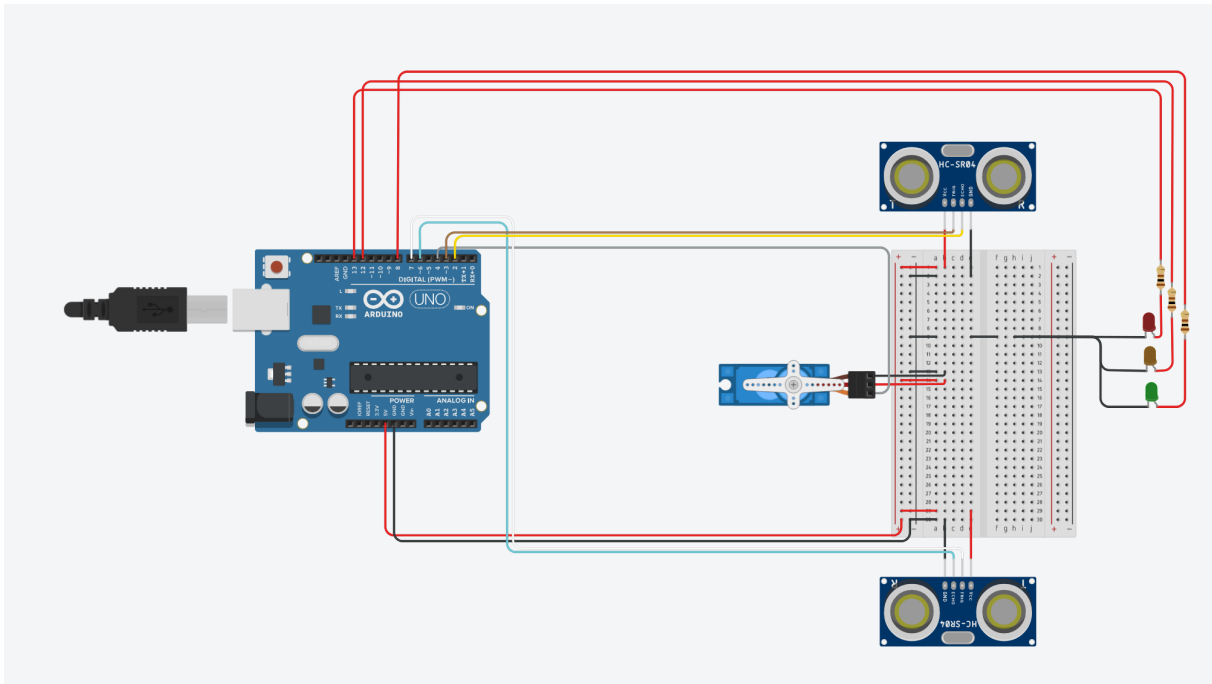


Figure 1: Tinkercad circuit of the automatic parking barrier system.

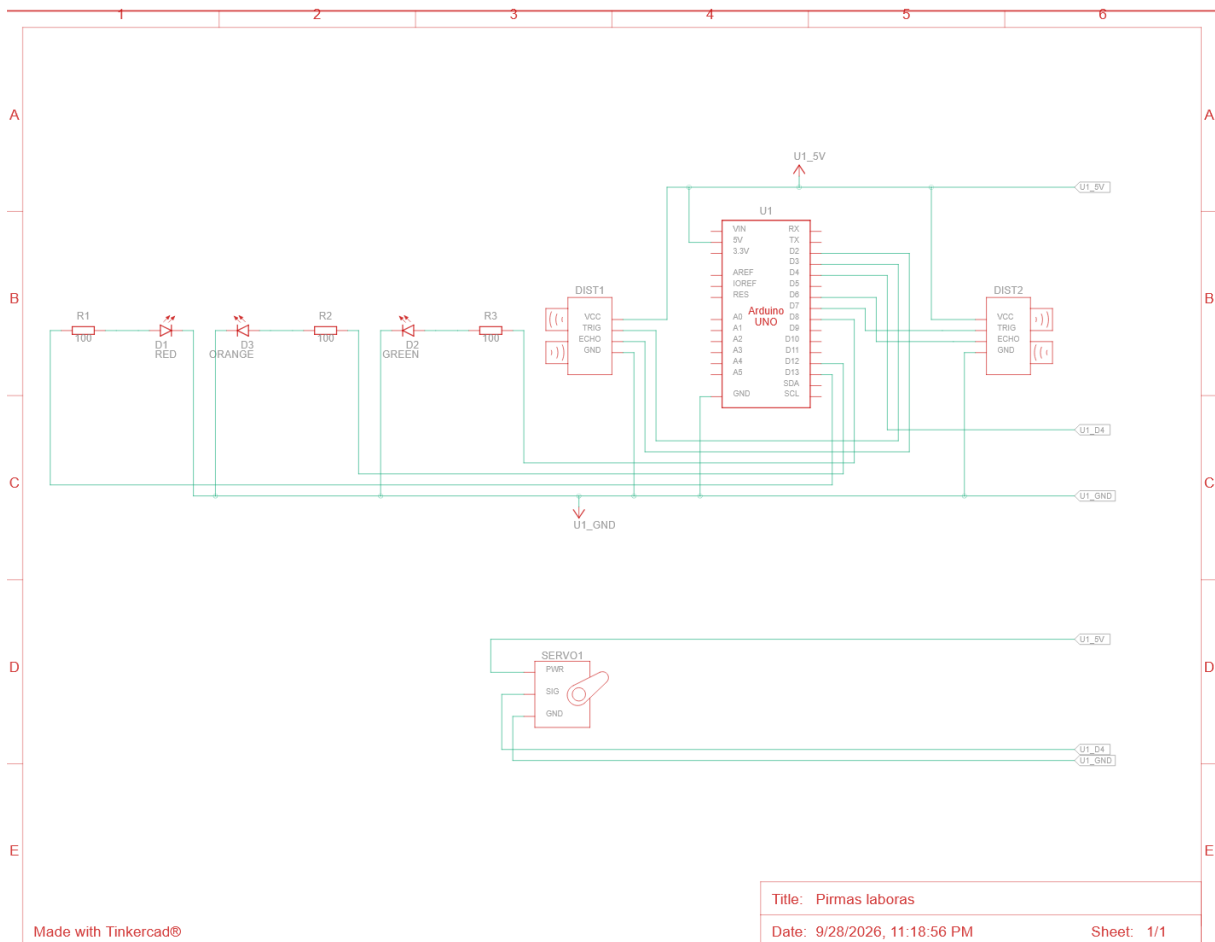


Figure 2: Tinkercad schematic of the circuit.

2 Results and Future Improvements

The main functionality of the project works correctly. The first ultrasonic sensor detects a vehicle within 100 cm and opens the barrier. After the vehicle passes the second sensor and moves away from it, the barrier closes again. The red and green LEDs correctly show whether the barrier is closed or open. The yellow LED turns on while the barrier is moving.

The original idea was to make the yellow LED flash while the barrier was opening or closing. This was not completed successfully, so the yellow LED currently stays on continuously while the servo motor is moving.

Possible future improvements include making the yellow LED flash while the barrier is moving and building a real physical miniature model of the parking barrier. Another improvement would be to add a license plate recognition system to identify vehicles entering and leaving the parking area. The system could also measure and store how long each vehicle stays inside the parking area.